PRD: Industrial Waste & By-Product Exchange Platform

Version: 1.0 (Draft) | **Status:** For review | **Stack:** React, CSS, FastAPI, SQLite (to PostgreSQL later)

1. Overview

Industries generate large amounts of leftover material, scrap and by-product. Much of it could be reused or recycled, but there is no trusted, structured way for the companies that produce it to find the companies that need it. As a result, usable material is wasted while other companies buy fresh raw material.

This platform is a B2B marketplace where companies list their waste, leftover material and by-products, and other companies (especially small and medium ones) buy them as raw material. Every company is verified through its business license, every purchase is checked for suitability, and every company's confidential data stays hidden from other users.

2. Problem Statement

- Producers have no easy way to find buyers for waste and by-products, so they discard or under-sell them.
- Buyers do not know which waste streams are usable for their process, and do not trust unverified sellers.
- The same material is named differently by different companies, so search and matching fail.
- Pricing is unclear: buyers cannot see the real landed cost (material + processing + transport).
- Companies fear sharing business data with competitors.

3. Goals and Non-Goals

Goals

1. Give companies a single place to list and discover reusable industrial waste and by-products.
2. Reduce demand for virgin raw material by redirecting waste into production.
3. Ensure only verified, genuinely suitable companies can buy a given material.
4. Make pricing transparent and fair to both sides (seller does not lose, buyer does not overpay).
5. Protect every company's confidential information.

Non-Goals (for the first version)

- Payment gateway and escrow (billing only for now; see Future Scope).
- Logistics booking or fleet management (only transport cost estimation).
- Mobile apps (responsive web only).
- International trade and customs.

4. Target Users

User	Description	Main need
Seller company	Any industry producing waste, leftover or by-product	Sell it instead of discarding it, at no loss
Buyer company	Usually small or medium manufacturer	Cheaper raw material that fits its process
Platform admin	Our internal team	Verify licenses, manage the material catalog, resolve disputes

5. Key Concepts

- **Material catalog:** A fixed list of material types controlled by the platform. Sellers choose from it; they cannot invent names.
- **Listing:** A seller's offer of a material, with total quantity, price and location.

- **Order / purchase request:** A buyer's request for some quantity of a listing; needs seller approval.
- **Compatibility check:** An automatic test of whether the material is useful for the buyer's actual products.
- **Cost bill:** Material cost + processing cost + transport cost for the requested quantity.
- **Unit:** All quantities are in **tons**, everywhere.

6. Features and Requirements

F1. Listing of Leftover Material, Waste and By-Product

Description: Verified sellers can create and manage listings.

Requirements

- F1.1 A seller can create a listing with: material type (from catalog), category (leftover / waste / by-product), total quantity (tons), price per ton, processed or raw, pickup location, availability dates and a description.
- F1.2 A seller can edit, pause, close or delete a listing (edit is restricted once an order is approved).
- F1.3 Listings show remaining quantity, updated automatically as orders are approved.
- F1.4 Buyers can search and filter by material type, quantity, price, distance, and availability.
- F1.5 Seller identity is shown only as an anonymous label (e.g. "Verified Seller #1042") and a general region, never the company name.

Acceptance criteria: A verified seller can publish a listing in under 3 minutes; an unverified seller cannot publish.

F2. Compatibility Check

Description: The platform checks whether the material genuinely fits the buyer's production, using the products declared in the buyer's license and profile.

Requirements

- F2.1 Each catalog material is mapped to the industries / product types that can use it (a compatibility map maintained by admins).
- F2.2 When a buyer views a listing, the system compares the material against the buyer's verified products.

- F2.3 **Compatible:** show a green "Suitable" badge.
- F2.4 **Not compatible:** show a red "Not suitable" badge on the listing.
- F2.5 If the buyer still tries to place an order, show a **second, blocking warning** ("This material is not suitable for your products") that requires explicit confirmation.
- F2.6 Log every override (buyer, material, time) so admins can review misuse; admins may restrict repeat offenders.

Open question: Should override be allowed at all, or only after admin approval? (See Section 14.)

F3. Economic Feasibility and Cost Bill

Description: The final price must not cause a loss to the seller and must not be unfair to the buyer. A bill is generated from three cost parts.

Cost components

1. **Actual material price:** seller's price per ton x requested tons.
2. **Processing price:** applies only if the material has been processed. The seller must declare the processing steps and cost; the platform cross-verifies it against reference ranges for that material and process. Claims far outside the range are flagged for admin review.
3. **Transport fee:** calculated from the distance between seller and buyer locations (per ton per km rate), plus toll charges where route data is available.

Requirements

- F3.1 Bill formula: `Total = (material price + processing price) x tons + transport fee`.
- F3.2 The seller sets a minimum acceptable price (floor) so a sale never goes below their cost; listings cannot be published below the verified cost floor.
- F3.3 The buyer sees a full itemized bill before placing the request.
- F3.4 Transport estimate shows distance, rate and toll (if available) as separate lines.
- F3.5 Processed-price verification results are visible to admins; buyers only see "Price verified".
- F3.6 The bill is recalculated automatically whenever quantity changes.

F4. Variable (Partial) Quantity

Description: A buyer can purchase any quantity up to what is available, not the whole lot.

Requirements

- F4.1 The buyer enters the required tons (e.g. 150 from a 1,000-ton listing). Cost is calculated only on the requested tons.
- F4.2 The seller can set an optional minimum order quantity and step size.
- F4.3 Every request needs **seller approval**. The seller can approve, reject or counter with a different quantity.
- F4.4 On approval, remaining quantity is reduced (1,000 to 850). Pending requests reserve quantity for a limited time (e.g. 48 hours) to avoid overselling.
- F4.5 If a request is rejected or expires, reserved quantity is released.

F5. Data Confidentiality

Description: Information collected at registration (license, production details, addresses, contacts, volumes) is private.

Requirements

- F5.1 Data collected at sign-up is never shown to other companies.
- F5.2 Counterparties see only anonymous labels, general region and verification status.
- F5.3 Exact address, contact details and company identity are revealed only after an order is approved and only to the extent needed to complete the pickup (decision needed; see Section 14).
- F5.4 License documents are stored encrypted, accessible only to admins and the verification service.
- F5.5 Compatibility and quantity checks run server-side; buyers get only a pass/fail result, never the underlying data.
- F5.6 Role-based access control and audit logs for all access to sensitive data.

F6. Standardized Material Types and Units

Requirements

- F6.1 Admins maintain the master material catalog (name, category, description, hazard class, compatible industries).
- F6.2 Sellers must select from the catalog. If a material is missing, they can submit a "request new material type" that admins review.
- F6.3 Quantity is always in **tons**. Sellers with other units must convert before entering; the UI offers a helper but stores tons only.

- F6.4 Decimal tons are supported (e.g. 0.5 ton).

F7. Coming Soon / Pre-Book

Description: Companies that know scrap will be generated by a certain date can list it in advance.

Requirements

- F7.1 A seller can create a "Coming Soon" listing with expected quantity and expected availability date.
- F7.2 Buyers can pre-book a quantity; seller approval is required, same as normal orders.
- F7.3 Pre-book quantity is capped at the expected quantity.
- F7.4 When the date arrives, the seller confirms actual quantity; the listing converts to a normal listing and pre-bookings are adjusted proportionally if actual quantity is lower.
- F7.5 Buyers get notifications for date changes, cancellations and conversion.
- F7.6 Sellers with repeated unfulfilled pre-books get a reliability warning.

7. License Verification and Anti-Misuse

The business license is the basis of trust on the platform.

- **Registration:** Company uploads license and provides business details. Admin (later, partly automated) verifies authenticity.
- **Data extracted from license:** industry type, products manufactured, permitted capacity (where available).
- **Seller check:** Quantity listed should be realistic for the seller's declared industry and capacity; outliers are flagged for review.
- **Buyer check:** Quantity requested should be realistic for the buyer's capacity and products. This stops hoarding and buying more than can be used, which would just create new waste.
- **Status:** Pending, Verified, Rejected, Suspended. Only Verified companies can trade.
- **Re-verification:** Periodic, and when a license expires.

8. User Flows

Seller: Register, upload license, get verified, create listing (choose catalog material, tons, price, processing details), bill preview and cost-floor validation, publish, receive request, approve / reject / counter, complete handover.

Buyer: Register, get verified, search or filter materials, see suitability badge, enter quantity, see itemized bill, send request, seller approves, complete purchase. If "Not suitable": red badge, then a second warning on checkout.

Admin: Review license, manage catalog and compatibility map, review flagged listings, overrides and disputes.

9. Notifications

In-app and email notifications for: verification result, new request, approval / rejection / counter, reservation expiry, pre-book conversion, and flagged activity.

10. Non-Functional Requirements

- **Security:** HTTPS, hashed passwords, JWT-based auth with role-based access, encrypted document storage, input validation, rate limiting.
- **Privacy:** Data minimization in all API responses; sensitive fields never sent to other companies' clients.
- **Performance:** Search and listing pages respond in under 2 seconds for up to 10,000 listings.
- **Scalability:** Use SQLAlchemy with Alembic migrations so the move from SQLite to PostgreSQL is a configuration change plus data migration.
- **Reliability:** Atomic quantity updates (transactions) to prevent overselling.
- **Usability:** Responsive, simple UI usable by non-technical staff.
- **Auditability:** Log all approvals, overrides, price changes and data access.

11. Technical Approach

Layer	Choice	Notes
Frontend	React + CSS	Component-based; role-based routes (seller, buyer, admin)
Backend	FastAPI (Python)	REST APIs, Pydantic validation, auto API docs
Database	SQLite now, PostgreSQL later	Use SQLAlchemy ORM and Alembic; avoid SQLite-only features
Auth	JWT	Roles: seller, buyer, admin (a company can be both buyer and seller)
File storage	Local encrypted storage now	Move to cloud object storage later
Distance / toll	Maps / routing API	Needed for transport fee; fallback to straight-line distance x factor

Core data entities

- **Company:** id, status, license data (private), industry, declared products, capacity, location
- **User:** id, company_id, role
- **Material (catalog):** id, name, category, hazard class, compatible industries
- **Listing:** id, seller_id, material_id, total_tons, remaining_tons, price_per_ton, processed flag, processing cost, min order, status (draft / live / coming_soon / closed), available_from
- **Order:** id, listing_id, buyer_id, tons, cost breakdown, status (pending / approved / rejected / expired / completed), is_prebook, compatibility_override flag
- **Audit log:** actor, action, target, timestamp

Core API groups

`/auth`, `/companies` (profile, license upload, verification), `/materials` (catalog), `/listings` (CRUD, search, coming soon), `/orders` (request, approve, reject, counter), `/bill` (cost estimate), `/admin` (verification, catalog, flags), `/notifications`.

12. Success Metrics

- Number of verified companies (sellers and buyers).
- Tons of waste listed and tons successfully sold (primary impact metric).
- Estimated virgin raw material avoided (tons).
- Listing-to-order conversion rate and approval rate.
- Percentage of orders flagged as unsuitable or over-quantity (should decrease over time).
- Average time from listing to sale.

13. Milestones (suggested)

Phase	Scope
1. Foundation	Auth, company registration, license upload, admin verification, material catalog
2. Core marketplace	Listings, search, partial-quantity orders, seller approval, itemized bill (material + transport)
3. Intelligence	Compatibility check and warnings, quantity sanity checks, processed-price verification
4. Advanced	Coming Soon / pre-book, notifications, audit and admin dashboards
5. Hardening	Move to PostgreSQL, security review, testing, pilot launch

14. Risks and Open Questions

Risks

- Processed-price verification needs reference cost data that may be hard to obtain; start with admin review and ranges.
- License data may not list products or capacity clearly; may need extra structured fields at sign-up.
- Hazardous waste has legal rules (handling, transport, disposal permits); needs compliance review.

- Cold start: marketplaces need both sellers and buyers; consider launching in one industry cluster or region first.
- Off-platform dealing once identities are revealed.

Open questions

1. Should buyers be allowed to override a "Not suitable" warning, or should it need admin approval?
2. When exactly are company identity and address revealed (after approval, after payment)?
3. Who sets the transport rate per ton per km, and how are tolls sourced?
4. Will the platform charge a commission or subscription?
5. Are hazardous materials in scope for version 1?
6. How are disputes (quality mismatch, weight differences) handled?

15. Future Scope

Online payments with escrow, logistics partner integration, quality certificates and lab reports, automated license verification via government databases, buyer wish-lists and demand posting ("I need X tons of Y"), analytics on carbon and waste savings, and mobile apps.